DOSE OF REALITY

Overlap Deceives, Exceedance Is Real

A Dietary Supplement Usage Analysis Using NHANES 1999-2023 Data

Workintech Data Science Bootcamp — Capstone Project #2 · Didem Arslan

46,388 person-cycle observations · 10 NHANES cycles · BigQuery + dbt + Python · svy (design-based statistics)

Research Question and Data

What happens when multiple supplements contain the same nutrient?

Overlap = Taking the same ingredient **from >1 different product** → **Exceedance** = Total intake exceeds the official **safe upper limit (UL)**

This project asks to what extent overlap actually leads to real dose exceedance.

DATA

Source:

CDC NHANES, 1999-2023 (10 cycles, 24 years)

Scope:

Demographic data + 30-day supplement use + product-ingredient reference table

Method:

BigQuery + dbt (ELT) · Python (pandas, svy design-based statistics)

Overview

Before the hypotheses: what does the overall picture show?

46,388

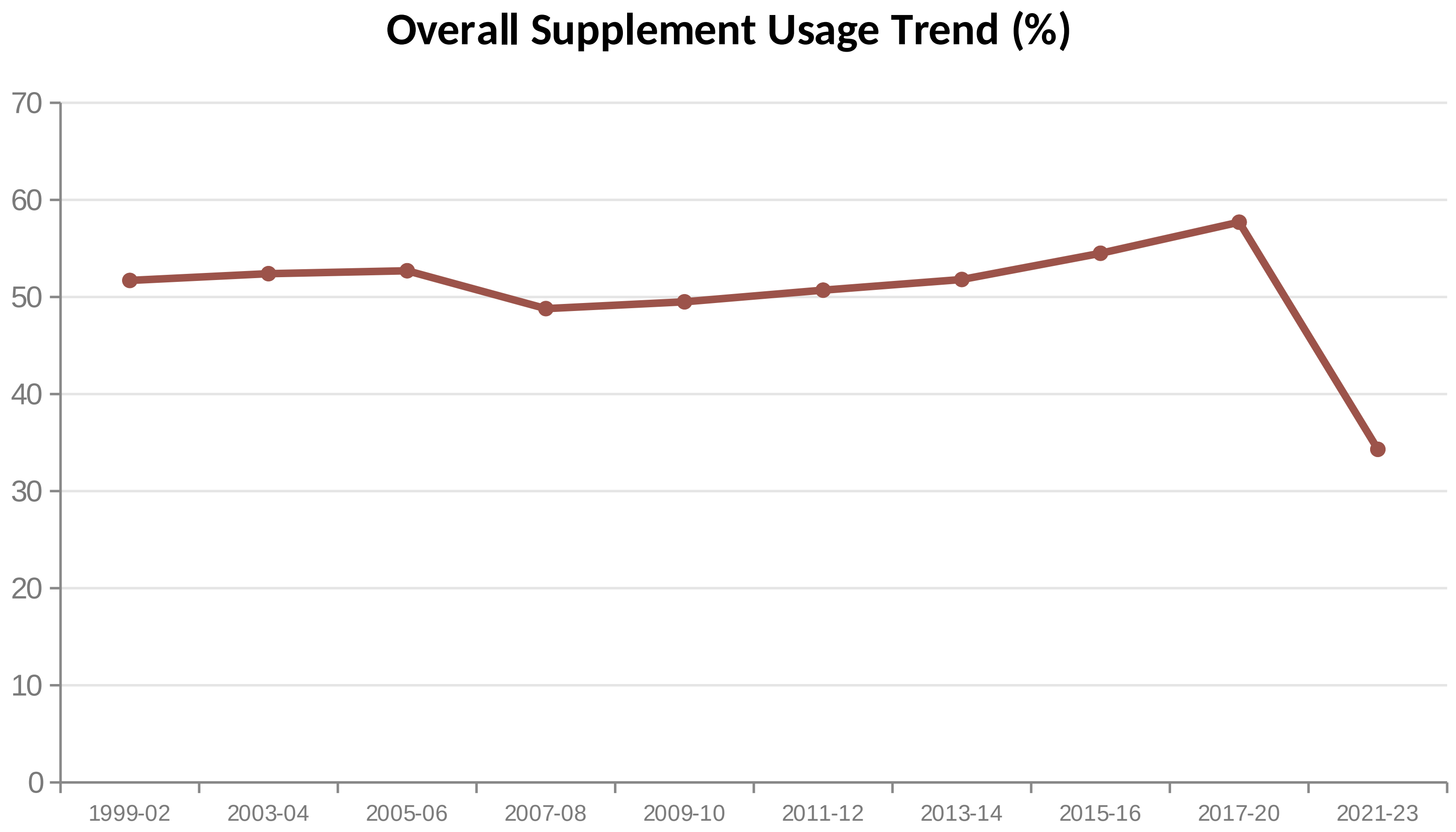
Person-cycle observations

10

NHANES cycles (1999-2023)

6

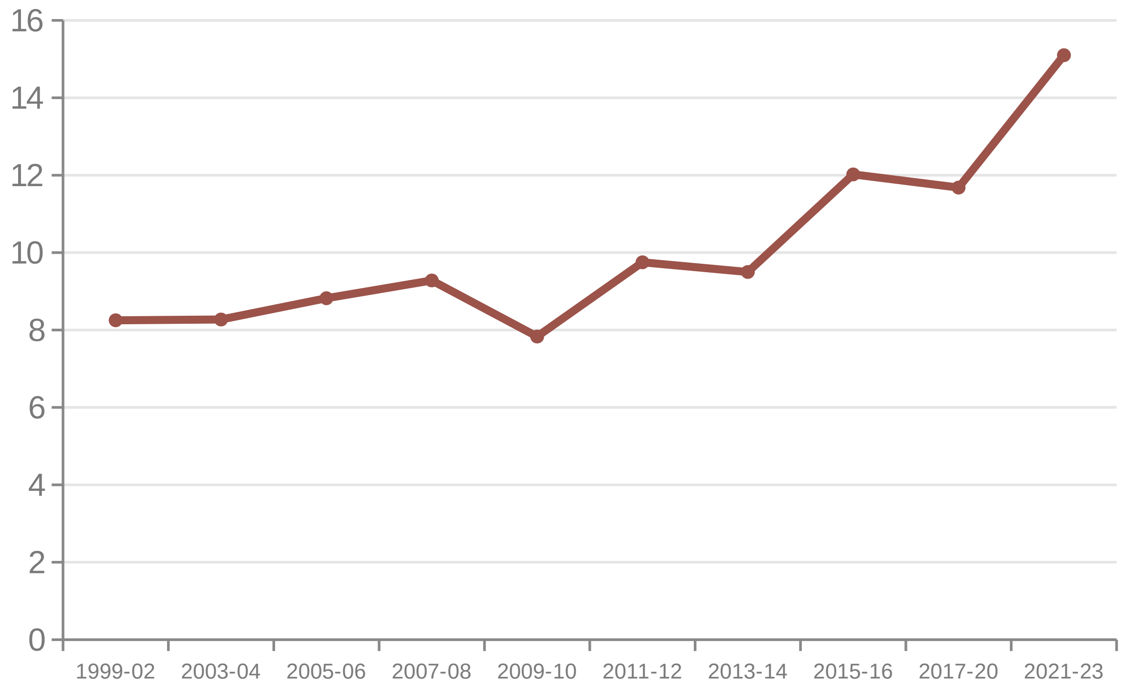
Ingredients examined



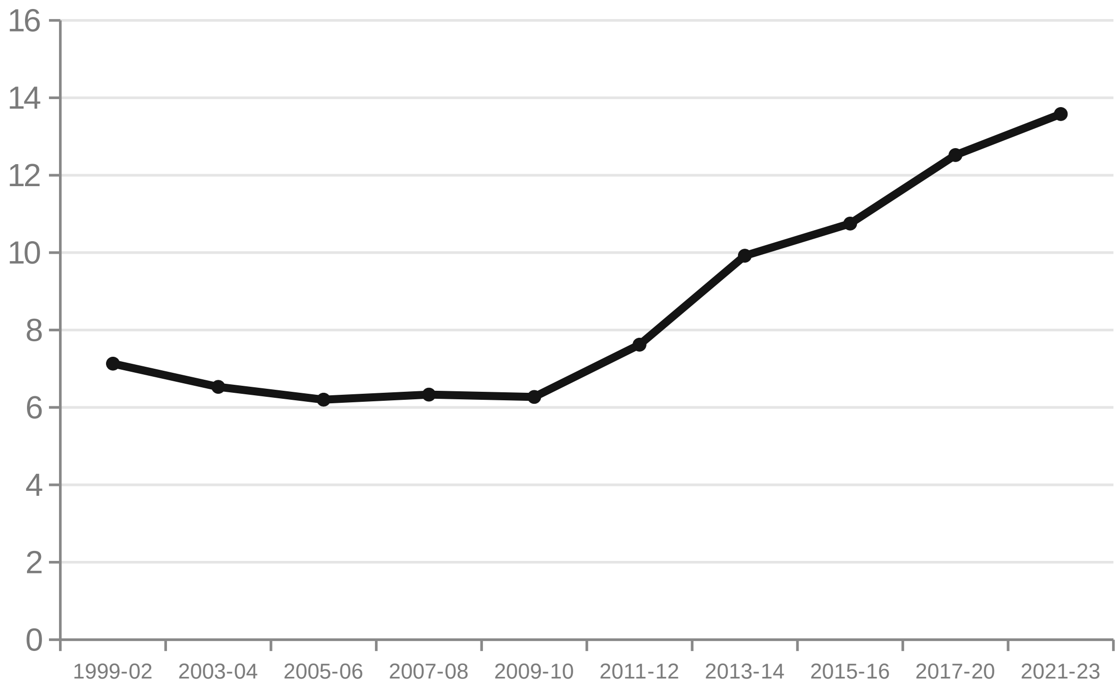
Overall Trend: As Usage Fell, Overlap and Exceedance Rose

Population-wide overlap and exceedance rates (average of 6 ingredients)

Overall Overlap Trend (%)



Overall UL Exceedance Trend (%)



While overall usage fell from 2017-2020 to 2021-2023 (57.7%→34.3%), overlap (11.7%→15.1%) and UL exceedance (12.5%→13.6%) kept rising — the remaining user base is behaving more intensively and riskily.

So What Does This Behavior Depend On?

Four questions, tested with four findings

H1 · Gender

Do overlap and exceedance differ between women and men?

H2 · Age Group

Does risk increase with age?

H3 · Ethnicity

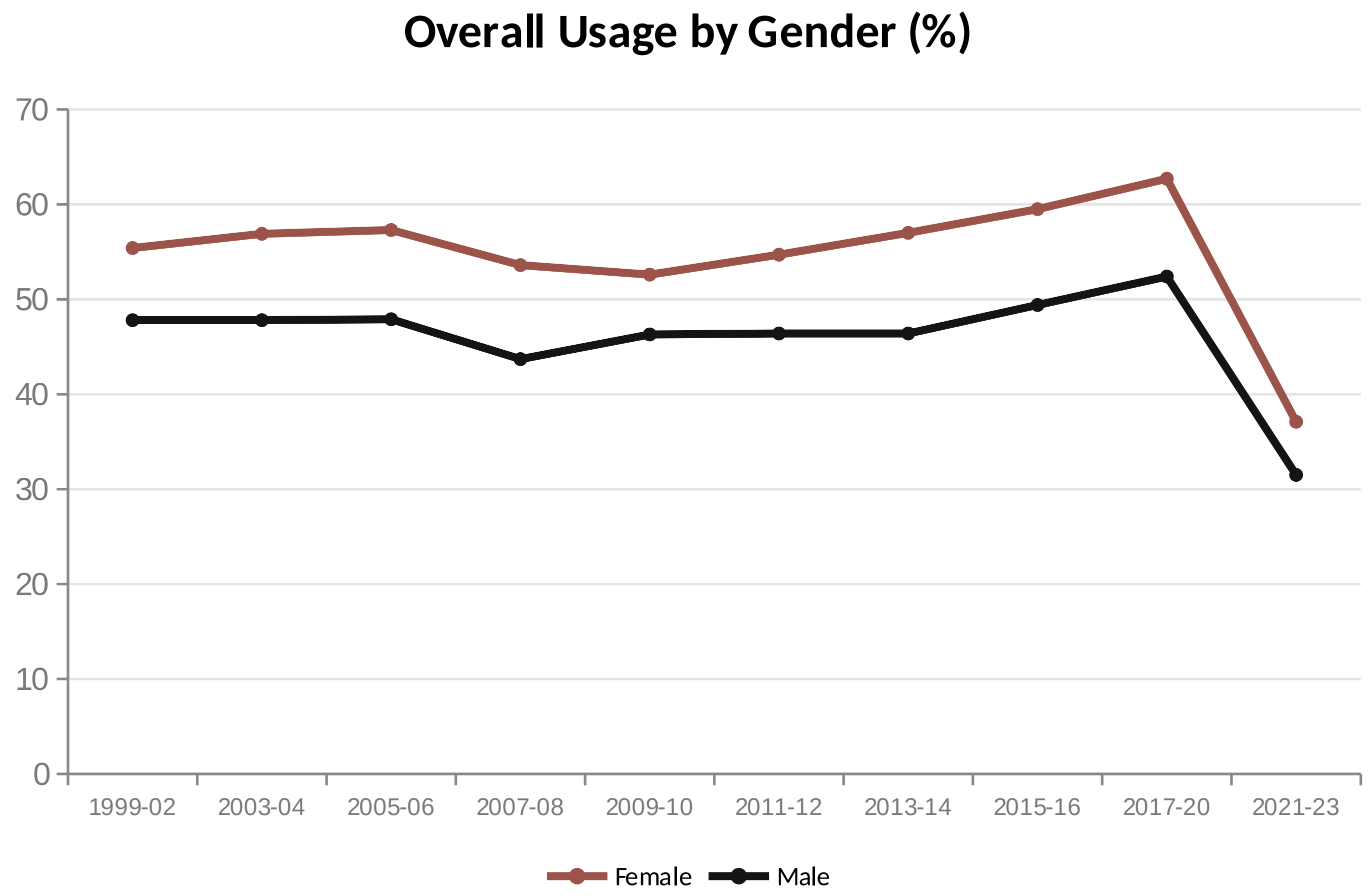
Which group stands out for which ingredient?

H4 · Pandemic Effect

How large is the 2021-2023 deviation, really?

H1 · Gender — Overall Trend

A consistent gap observed across 24 years



GENERAL OBSERVATION

Women consistently show higher usage rates than men across 24 years — the gap has been structural and persistent since 1999 (in the 7.6-10.3 point range).

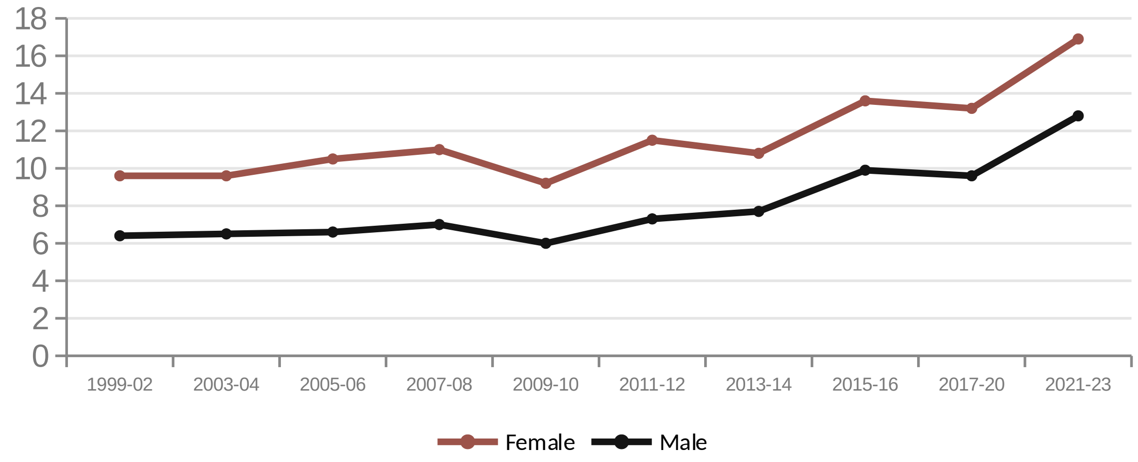
NEXT QUESTION

Does this trend hold similarly for overlap and dose exceedance?

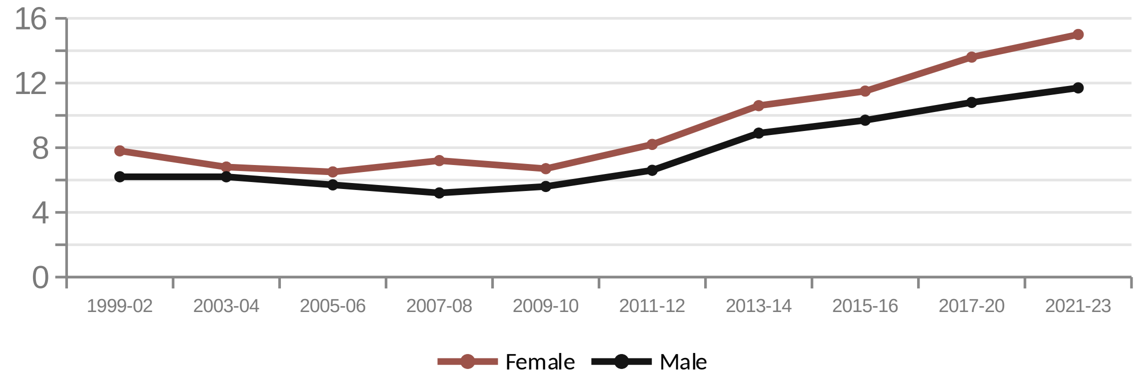
H1 · Gender — Overlap and UL Exceedance

A consistent gap observed across 24 years

Overlap (%)



UL Exceedance (%)



GENERAL OBSERVATION

Women consistently show higher overlap and exceedance rates than men across 24 years — the gap is structural and persistent.

NEXT QUESTION

Is this gap statistically significant, or just chance?

H1 · Gender — From Raw Finding to Verified Result

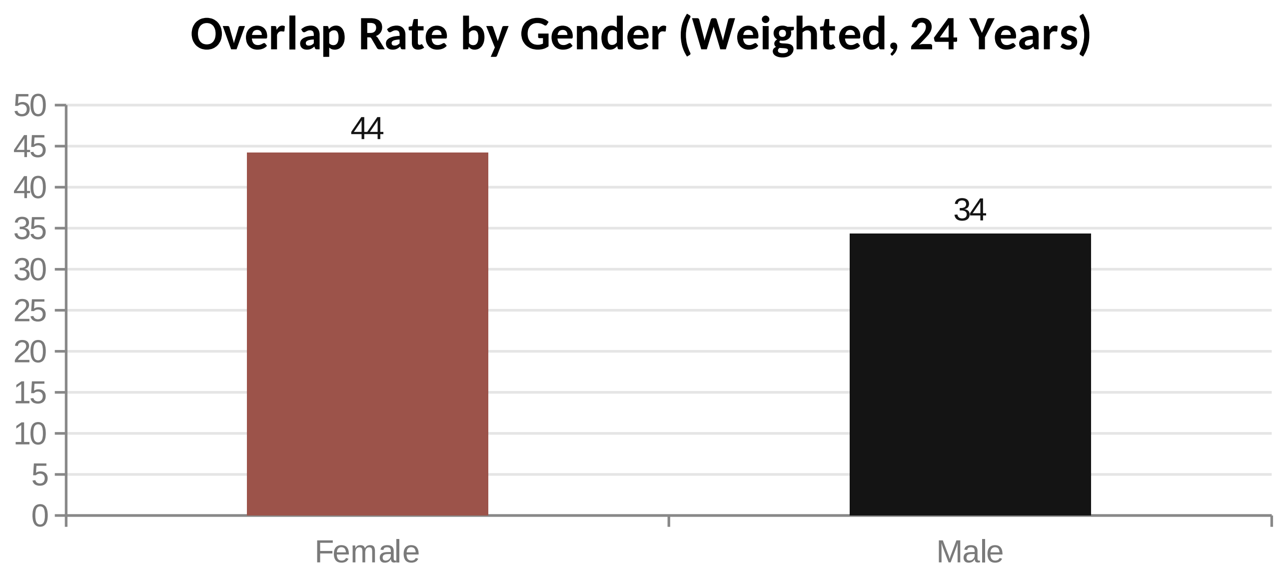
Applying NHANES's official weighting rule didn't change the result — it strengthened it

RAW FINDING

Overlap: 44.7% in women vs. 35.1% in men — unweighted rate.

WEIGHTED (OFFICIAL RULE)

Female 44.2% vs. Male 34.3% — the result held almost unchanged, confirmed.



Design-Based T-Test

Test statistic (t):

−13.76

Degrees of freedom (df):

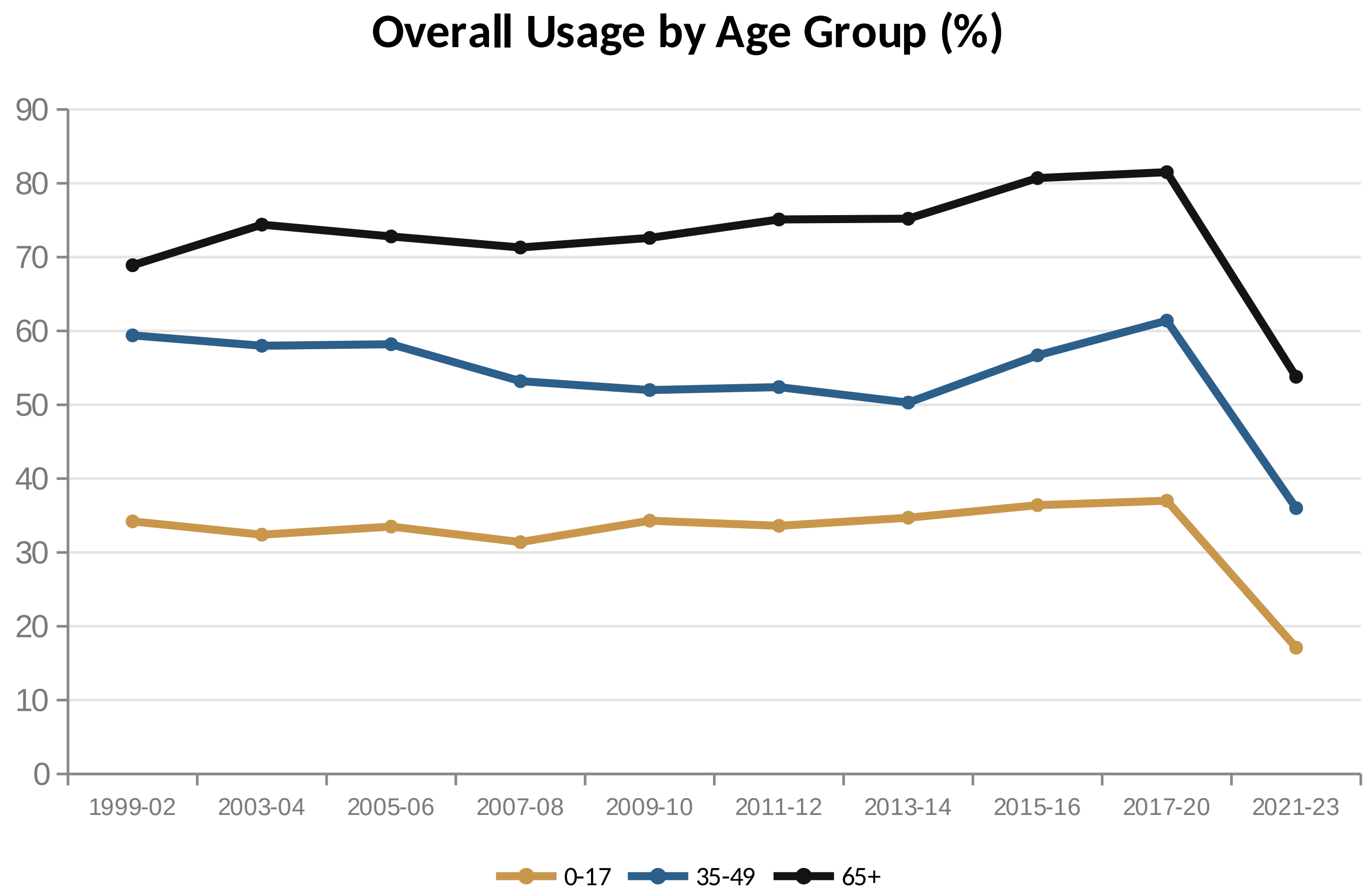
177

p-value:

< 0.0001 — Significant

H2 · Age Group — The Strongest Factor

Usage, overlap, and exceedance all rise monotonically with age



GENERAL OBSERVATION

The 65+ group shows the highest usage rate in every cycle. This monotonic pattern became clear from 2011 onward; before that (1999-2010) the ranking was more volatile.

Kruskal-Wallis Test

F-statistic:

584.79 (overlap)

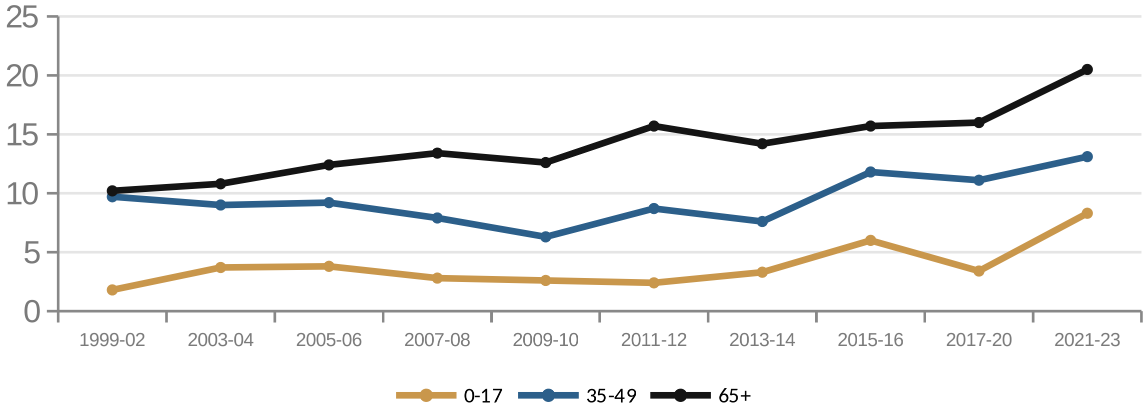
p-value:

< 0.0001 — Strongest factor

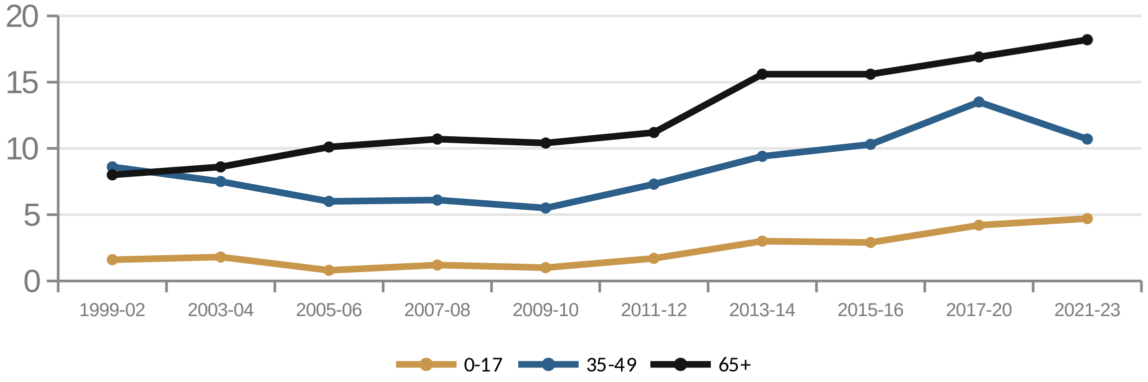
H2 · Age Group — Overlap and UL Exceedance

The strongest factor: risk rises monotonically with age

Overlap (%)



UL Exceedance (%)



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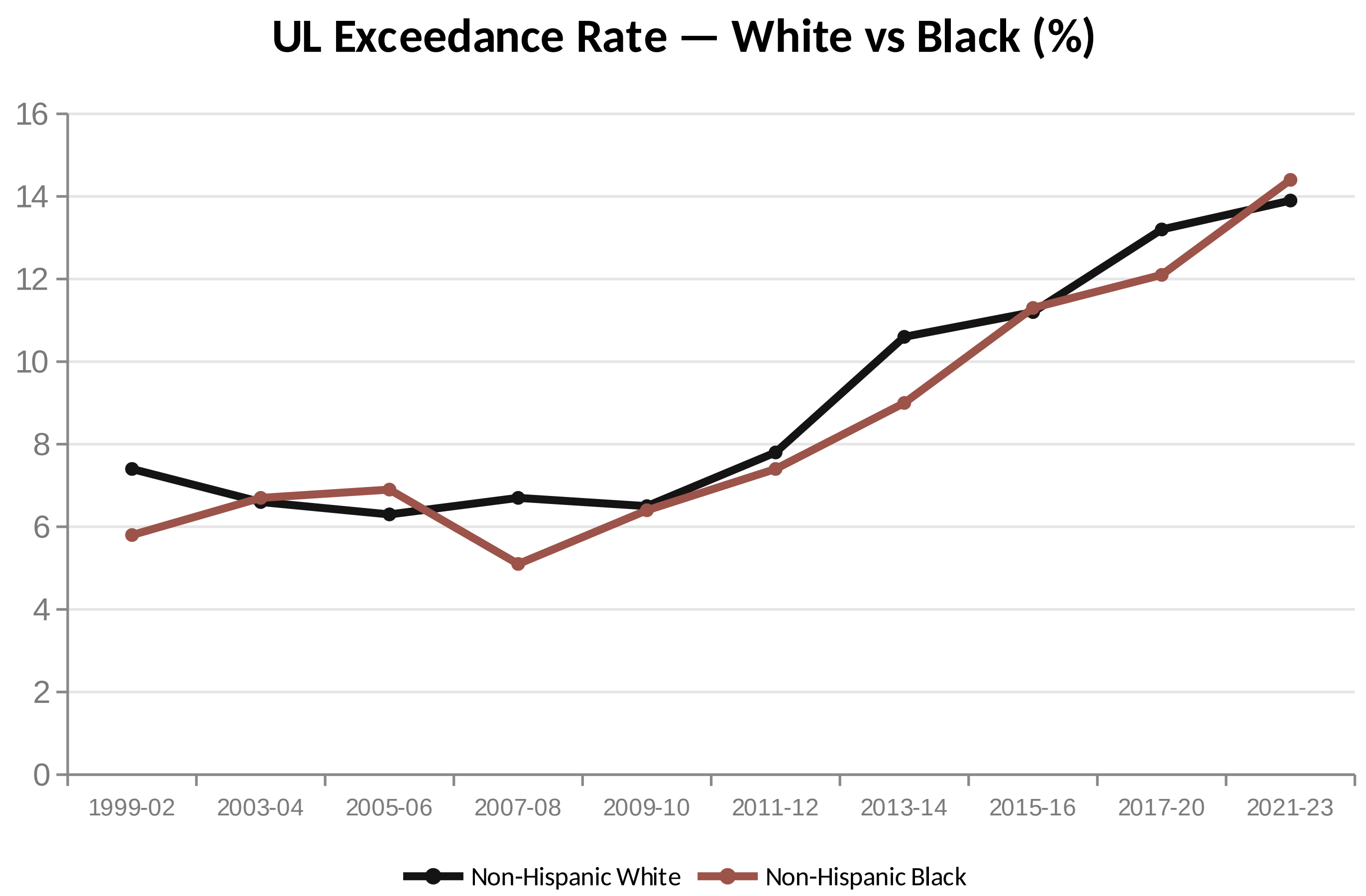
H3 · Ethnicity — A Shifting Hierarchy

The Non-Hispanic Black group overtook the White group in UL exceedance in 2021-2023



NOTABLE FINDING

2015-2016'ya kadar Black grubu hep gerideyken, 2021-2023'te (özellikle Demir'de) White grubunu geçti. NCHS'in bildirdiği anemi prevalansı eşitsizliğiyle (Black kadınlarda %31.4 vs White'ta %8.3) zamanlama olarak tutarlı.

Kruskal-Wallis Test

F-statistic:

12.69 (weakest, but significant)

p-value:

< 0.0001

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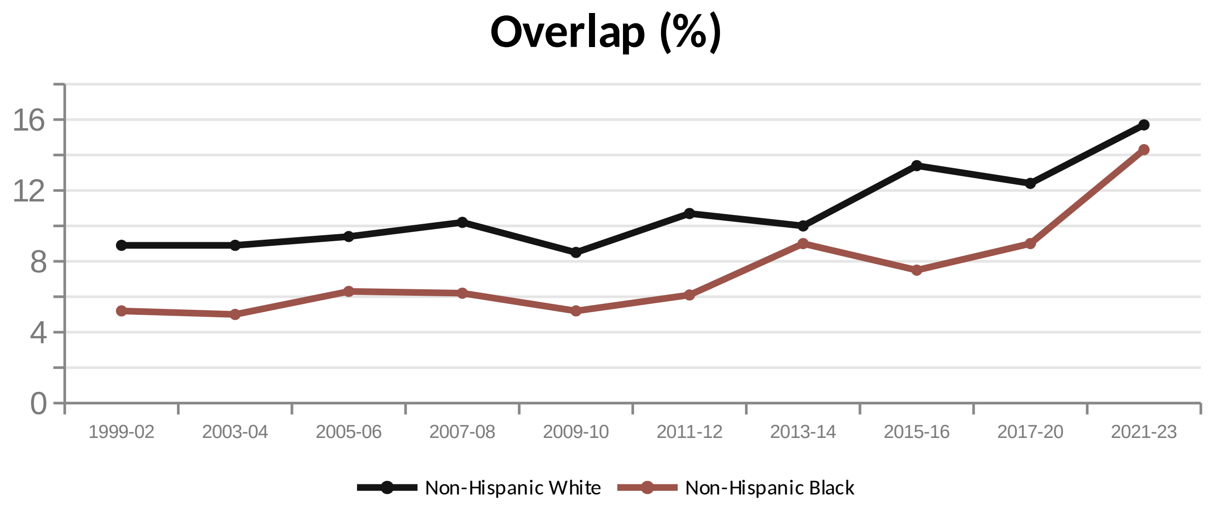
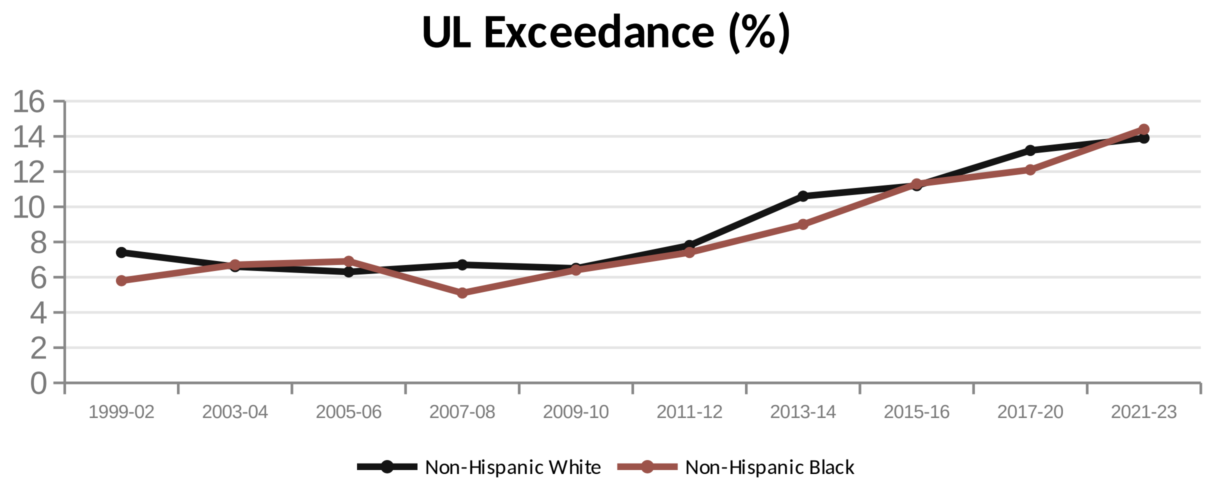




NOTABLE FINDING

Until 2015-2016 the Black group had consistently lagged behind; in 2021-2023 (especially for Iron) it overtook the White group.

Kruskal-Wallis Test

F-statistic:

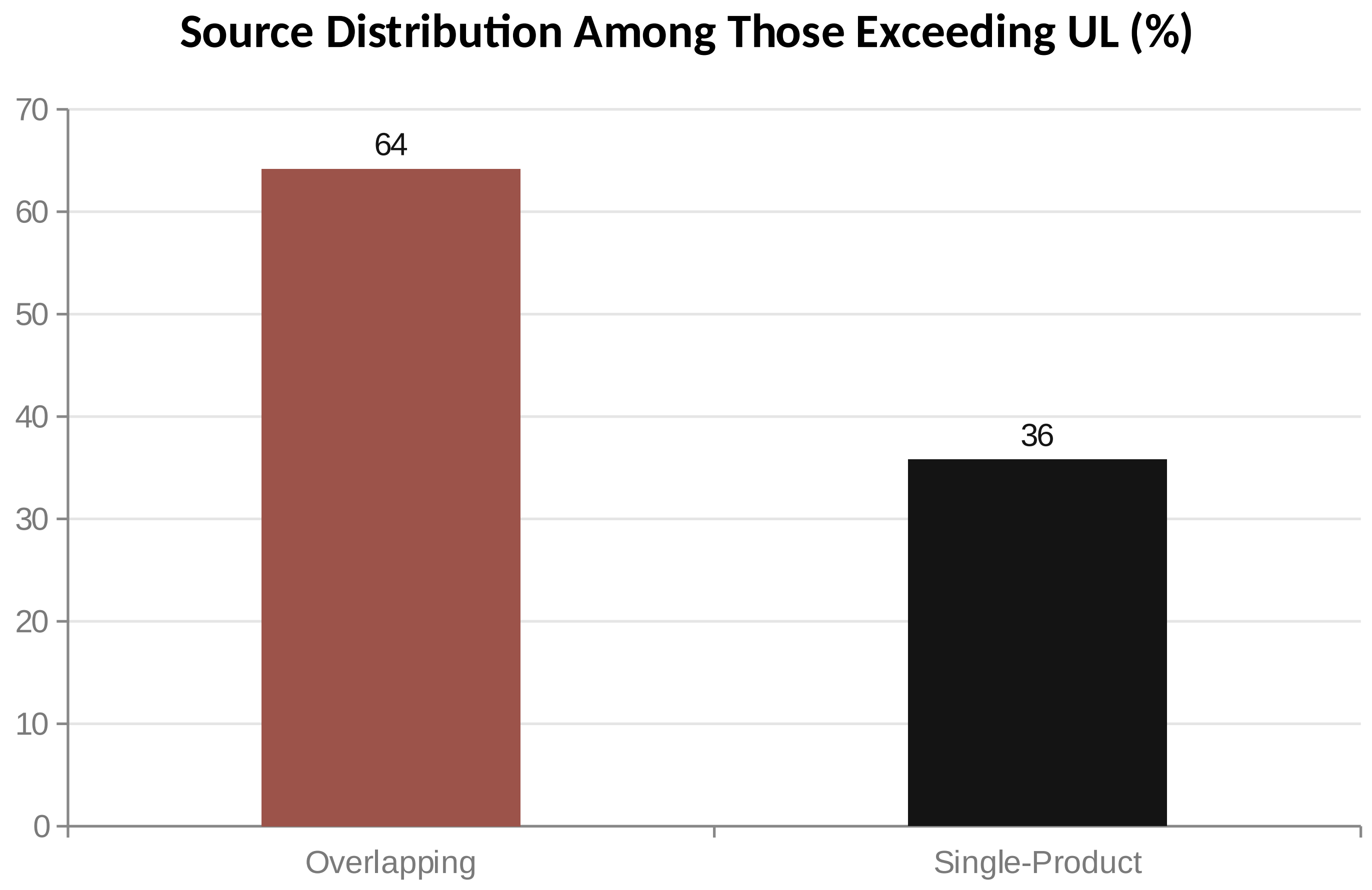
12.69 (weakest, but significant)

p-value:

< 0.0001

Source of Exceedance — Overlap or Single Product?

Across the population, exceedance stems systematically from overlap — not by chance



GENERAL OBSERVATION

Each ingredient carries a different mechanism: for Vitamin D the risk comes from overlap; for Iron and Niacin it comes from single-product dosage.

One-Sample Test (against 50%)

t-statistic:

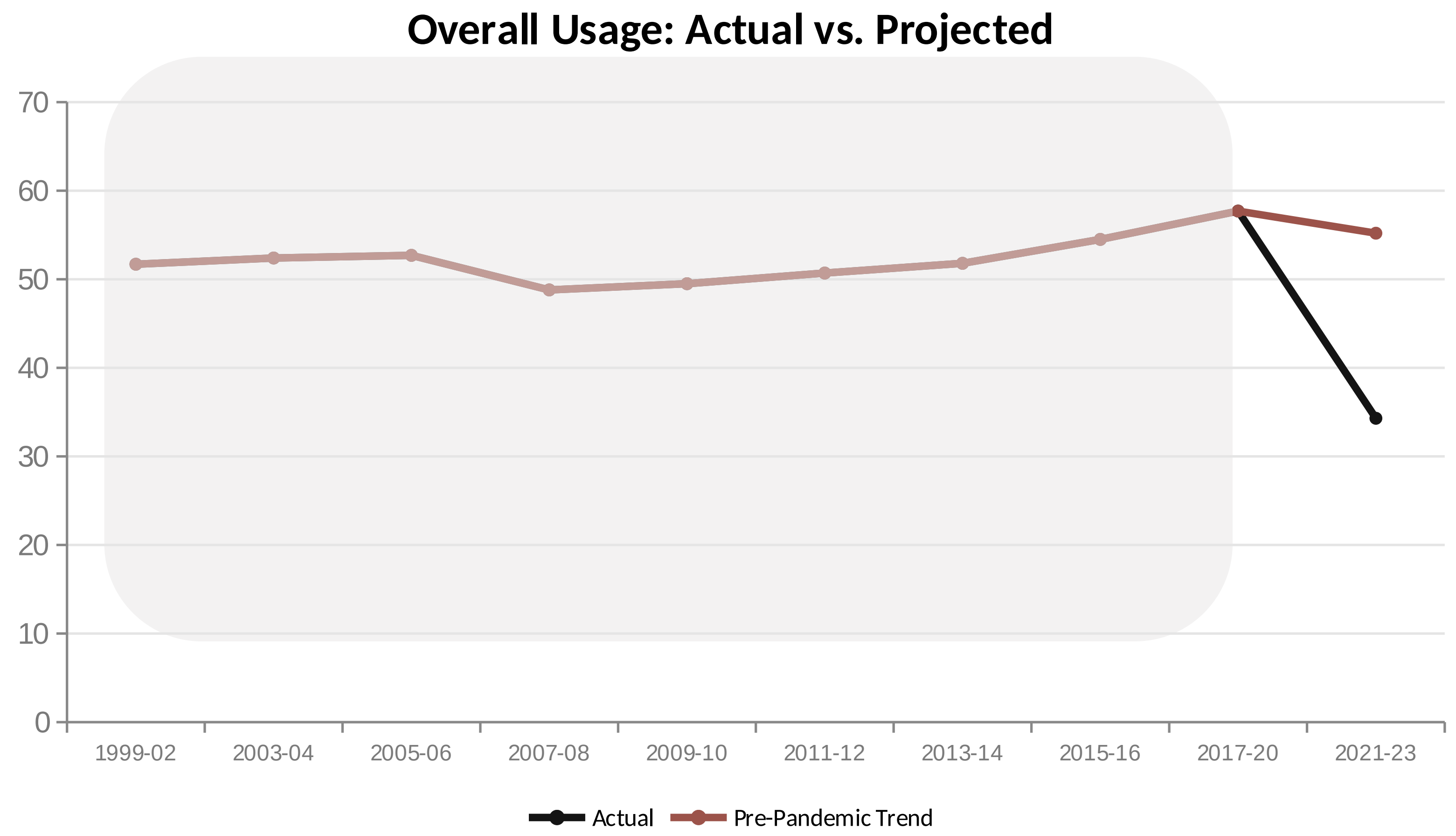
21.37

p-value:

< 0.0001 — Not by chance

Counterfactual Analysis — Pinning Down the Pandemic Effect

Instead of forecasting the future directly: 'what if the trend had continued?'



−20.9 puan

Confirmed deviation

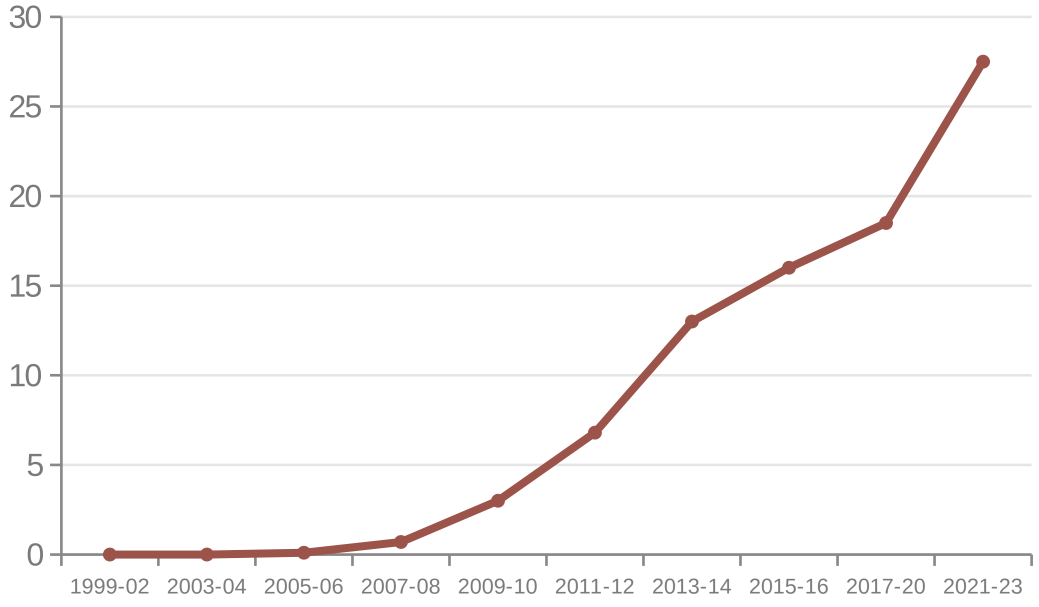
Interpretation

For Vitamin D and Zinc, exceedance came in ABOVE expectations — despite falling usage, the remaining user base intensified.

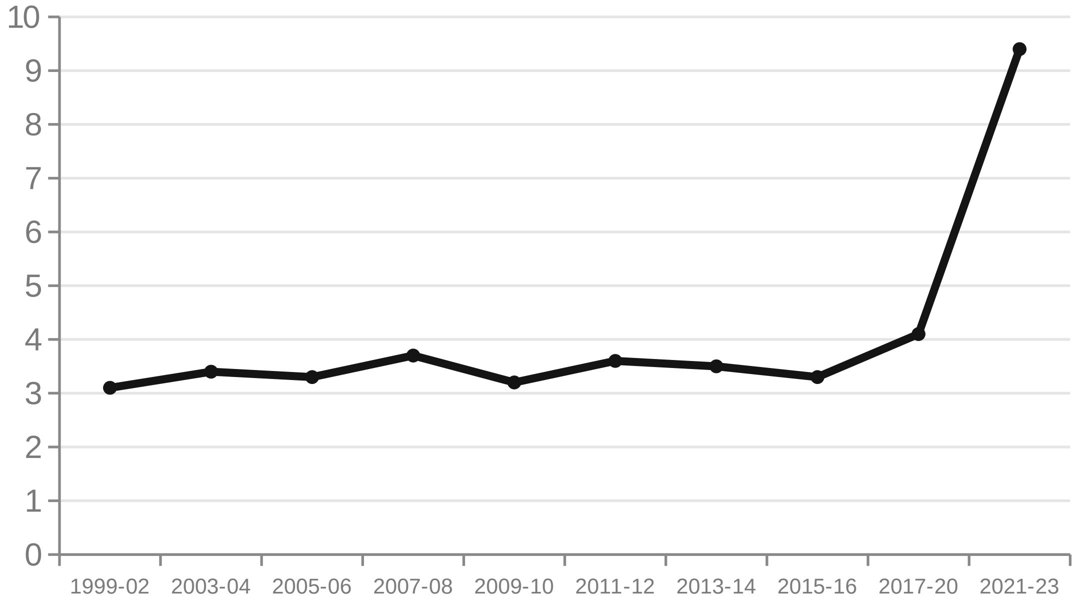
Vitamin D vs. Zinc — Two Different Risk Stories

Same apparent 'pandemic jump', but the underlying mechanisms are polar opposites

Vitamin D: Accelerating Momentum



Zinc: Sudden Surge From Zero

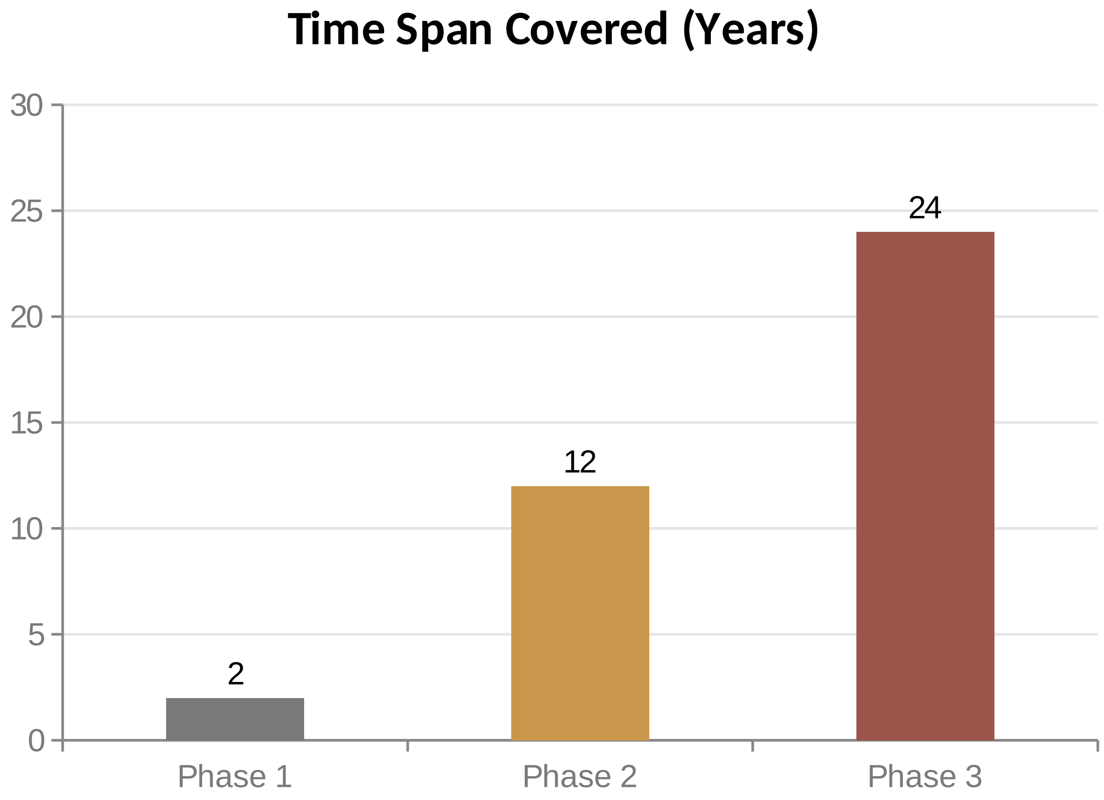


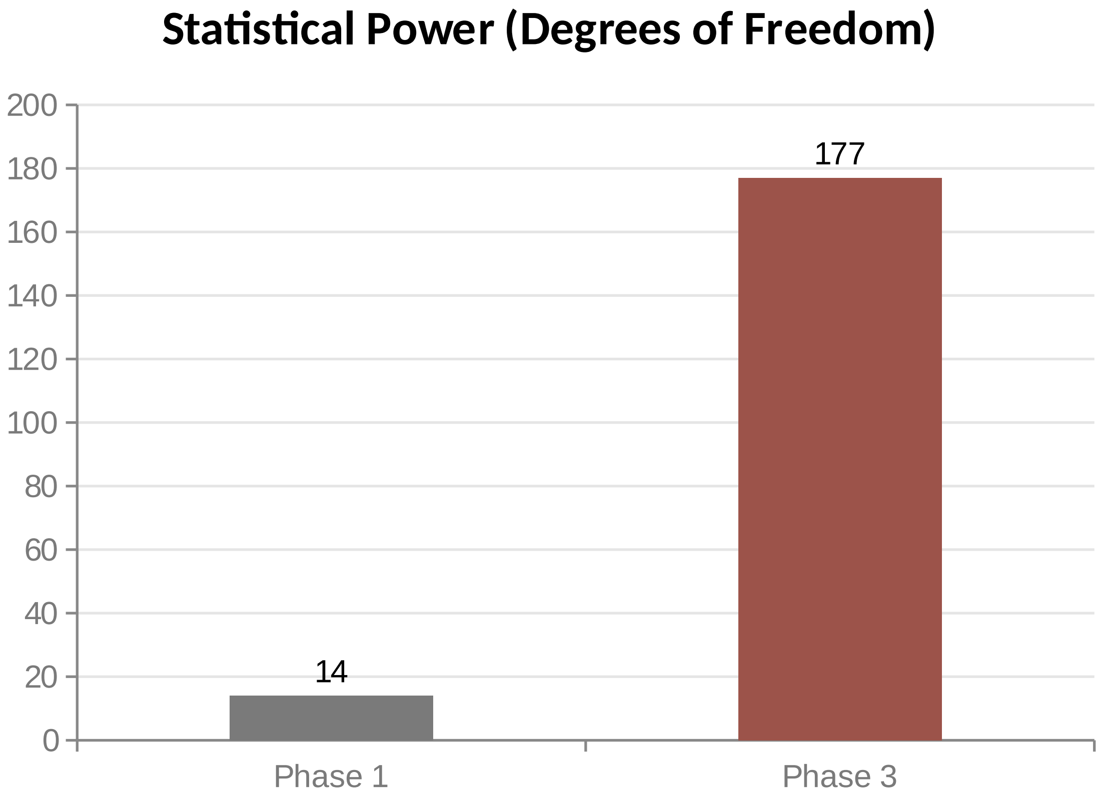
Vitamin D: a gradual, uninterrupted rise since low-awareness 1999-2006. Zinc: flat for 20+ years, with a sudden jump only after the pandemic.

Three-Phase Incremental Expansion

Each phase confirmed the previous one and added new findings

Phase 1 = 2021-2023 (single cycle) • Phase 2 = 2011-2023 (5 cycles) • Phase 3 = 1999-2023 (10 cycles)





Statistical Significance Tests — Summary

7 of 8 tests at p<0.0001, all design-based (svy)

Gender × Overlap / UL Exceedance

$t = -13.76 / -9.25 \rightarrow p < 0.0001$

Age Group × Overlap / UL Exceedance

$F = 584.79 / 531.62 \rightarrow p < 0.0001$

Ethnicity × Overlap / UL Exceedance

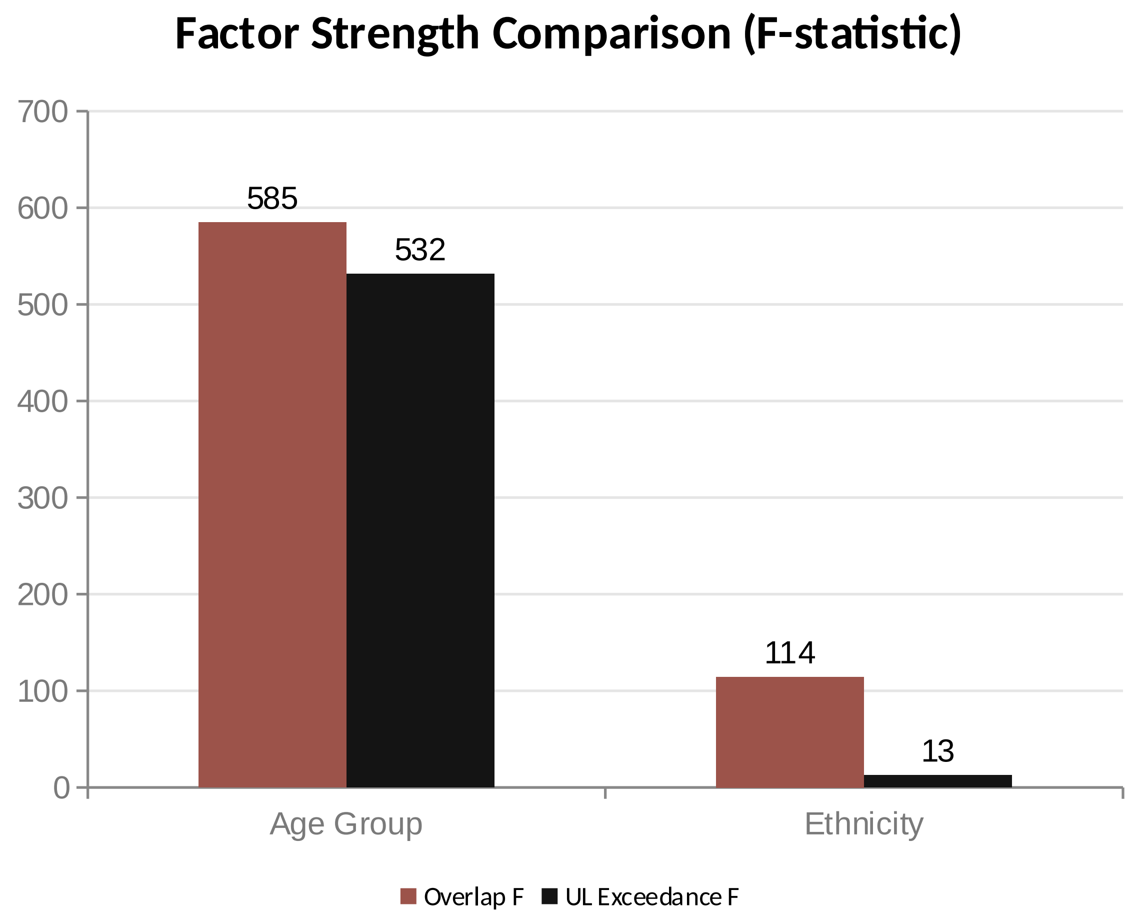
$F = 114.37 / 12.69 \rightarrow p < 0.0001$

Overlapping vs. Single-Product (against 50%)

$t = 21.37 \rightarrow p < 0.0001$

Iron × Targeted Subgroup (n=15)

$t = 1.33 \rightarrow p = 0.2049$ (not significant)



Conclusion

— Overall usage rose from 1999-2020 and fell in 2021-2023 — yet overlap and exceedance trends kept rising; most of the exceedance stems from overlap.

— 64.2% of exceedance comes from overlap — statistically proven ($p<0.0001$).

— Gender, age, and ethnicity effects are strong and significant across 24 years (10/11 tests at $p<0.0001$).

— Each ingredient carries its own risk mechanism — a single message isn't enough.

Increased usage does not mean more conscious use — using multiple products raises the risk of exceedance.

Detailed methodology, limitations, and full findings are available in the accompanying report and dashboard.

Thank you — Questions?